ISyE 6669 HW 13

Fall 2025

Note: There may be more than one correct way to model constraints in this homework. Verify that the formulation models the logical relations correctly and are linear (with integer variables).

1. Write linear constraints with continuous and integer variables for the following problems. You need to clearly define the variables that you introduce and give an explanation of your constraints.

   (a) If we invest \$50 or more on project 1, then we can only invest at most \$100 on project 2. Suppose the investment amount on each project is a continuous variable.

   Solution:

   Let x_1, x_2 be the amount invested in projects 1 and 2 respectively. Let M be the budget (or a large number if a budget does not exist). Then we have the following constraints:

$$z \geq \frac{x_1 - (50 - \varepsilon)}{(M - 50) + \varepsilon} \tag{1a}$$

$$x_2 \leq M(1 - z) + 100z \tag{1b}$$

   where ε is a small positive number, $z \in \{0, 1\}$, and $x_1, x_2 \geq 0$.

   Constraint (1a) introduces a binary variable z that gets set to 1 if $x_1 \geq 50$. Constraint (1b) limits x_2 to be at most \$100 if $x_1 \geq 50$ and at most M otherwise.

   (b) If two continuous variables x_1, x_2 satisfy the linear constraint $x_1 + x_2 < 1$, then they also need to satisfy $x_1 - x_2 \geq 1$. Plot the feasible region of (x_1, x_2) that satisfies the above if-then statement. Is this feasible region a convex set? Now use linear constraints with continuous and integer variables to model this if-then statement. Define all the variables you need and explain the constraints that you write down.

   Solution: We can see from Figure 1 that the feasible region is not convex, since we can select two points in the feasible region such that the line connecting the two points is not entirely contained in the feasible region. We model this if-then statement with the following constraints:

$$x_1 + x_2 \geq 1 - Mz \tag{2a}$$

$$x_1 - x_2 \geq 1 - M(1 - z) \tag{2b}$$

   where M is a large number, $z \in \{0, 1\}$, and $x_1, x_2 \in \mathbb{R}$. Constraint (2a) introduces a binary variable z that gets set to 1 if $x_1 + x_2 < 1$. If $x_1 + x_2 \geq 1$, then z is free to take on any value. Constraint (2b) ensures that $x_1 - x_2 \geq 1$ if $z = 1$, i.e., if $x_1 + x_2 < 1$.

2. Suppose that you are interested in choosing a set of investments $\{1, \ldots, 7\}$. Assume that the investments have cost of \$5, \$7, \$6, \$3, \$9, \$12, \$5. Assume there is a budget of \$30. Also assume that the eventual expected profit out of each investment is \$1, \$3, \$2, \$4, \$1, \$5, \$4. Write an optimization model to maximize profit with the following constraints. The objective function and all the constraints should be linear and there may be integer variables. Define all variables you need and explain the constraints that you write down.

   (a) You must choose at least one of the investments.

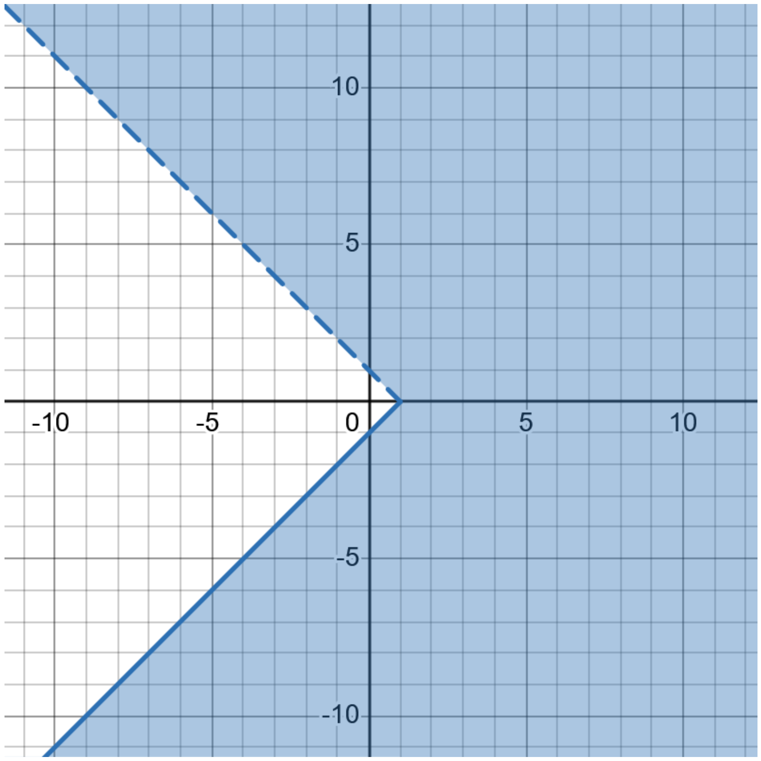

Figure 1: Feasible region of (x_1, x_2).

(b) Investment 1 cannot be chosen if investment 3 is chosen.

(c) Investment 4 can be chosen only if investment 2 is also chosen.

(d) You must choose either both investments 1 and 5 or neither.

(e) You must choose either at least one of the investments 1, 2, 3 or at least two investment from 2, 4, 5, 6.

Solution: Let $x_i \in \{0,1\}, i = 1,2,...,7$, denote whether to invest in project i (1 for investing, 0 for not). Let $y_1 \in \{0,1\}$ denote whether at least one of investments 1, 2, or 3 has been chosen and $y_2 \in \{0,1\}$ denote whether at least two investments from 2, 4, 5, or 6 have been chosen. Denote the costs as $c_1 = 5, c_2 = 7, c_3 = 6, c_4 = 3, c_5 = 9, c_6 = 12$, and $c_7 = 5$. Denote the expected profits as $p_1 = 1, p_2 = 3, p_3 = 2, p_4 = 4, p_5 = 1, p_6 = 5$, and $p_7 = 4$. Denote the budget as $B = 30$. The binary linear program can be formulated as follows:

$$
\begin{aligned}
\max \quad & \sum_{i=1}^{7} p_i x_i \\
\text{s.t.} \quad & \sum_{i=1}^{7} c_i x_i \le B \\
& \sum_{i=1}^{7} x_i \ge 1 \\
& x_1 + x_3 \le 1 \\
& x_4 \le x_2 \\
& x_1 - x_5 = 0 \\
& x_1 + x_2 + x_3 \ge y_1 \\
& x_2 + x_4 + x_5 + x_6 \ge 2y_2 \\
& y_1 + y_2 \ge 1 \\
& x_i \in \{0,1\}, \quad i = 1,2,\ldots,7 \\
& y_i \in \{0,1\}, \quad i = 1,2.
\end{aligned}
$$

The formulation need not be unique. The constraints are broken down further per question below. All variables are defined above.

(a) You must choose at least one of the investments.
$\sum_{i=1}^{7} x_i \ge 1$ This inequality will ensure that at least one x_i is equal to 1, thus making sure that at least one investment is chosen.

(b) Investment 1 cannot be chosen if investment 3 is chosen.
$x_1 + x_3 \le 1$
If $x_1 = 1$, then above inequality will forcce $x_3 = 0$. However if $x_1 = 0$ then above becomes $x_3 \le 1$ thus not limiting x_3 in any way.

(c) Investment 4 can be chosen only if investment 2 is also chosen.
$x_4 \le x_2$
Basically we want to enforce that if 2 is not chosen then 4 is also not chosen. If $x_2 = 0$ then above inequlaity will force $x_4 = 0$, and if $x_2 = 1$ then above inequality will become $x_4 \le 1$, thus not restricting x_4.

(d) You must choose either both investments 1 and 5 or neither.
$x_1 - x_5 = 0$
This will ensure that both x_1 and x_5 have same value thus ensuring that either both 1 and 5 are chosen or they are both not chosen.

(e) You must choose either at least one of the investments 1, 2, 3 or at least two investment from 2, 4, 5, 6.
$x_1 + x_2 + x_3 \ge y_1$
$x_2 + x_4 + x_5 + x_6 \ge 2y_2$
$y_1 + y_2 \ge 1$

3. To graduate from ABC University with a major in Analytics, a student must complete at least two math courses, at least two OR courses, and atleast two computer courses. Some courses can be used to fulfill more than one requirement.

(a) Calculus can fulfill the math requirement;

(b) Optimization, math and OR requirements;

(c) Data structures, computer and math requirements;
(d) Computer simulation, OR and computer requirement;
(e) Introduction to computer programming, computer requirement;
(f) Forecasting, OR and math requirement; and
(g) Business Statistics, OR requirement.

Some courses are pre requisites for others:

(a) Calculus is a prerequisite for Business Statistics.
(b) Introduction to computer programming is a prerequisite for Computer simulation and Data structures
(c) Business Statistics is a prerequisite for Forecasting.

Formulate an integer program to minimize the number of courses needed to satisfy the major requirements.

Solution: Let $x_C, x_O, x_{DS}, x_{CS}, x_{ICP}, x_F, x_{BS}$ represent variable indicating whether certain class is taken on not, i.e. x_C is variable indicating whether Calculus is taken or not, etc. Then we can formulate problem as:

$$\begin{aligned}
\min \quad & x_C + x_O + x_{DS} + x_{CS} + x_{ICP} + x_F + x_{BS} \\
\text{s.t.} \quad & x_O + x_{CS} + x_F + x_{BS} \geq 2 \\
& x_C + x_O + x_{DS} + x_F \geq 2 \\
& x_{DS} + x_{CS} + x_{ICP} \geq 2 \\
& x_{BS} \leq x_C \\
& x_{CS} \leq x_{ICP} \\
& x_{DS} \leq x_{ICP} \\
& x_F \leq x_{BS} \\
& x_C, x_O, x_{DS}, x_{CS}, x_{ICP}, x_F, x_{BS} \in \{0,1\}
\end{aligned}$$

where the first three constraints are ensuring that program requirements are met and next four are ensuring that prerequisits are met.

4. A machine tool plant owns four different machines on which it can process jobs. If a machine is used at all, then a setup time is needed. A job cannot be divided between machines, that is each job must be processed by exactly one machine. (Note however that one machine can process more than one job). The relevant times in minutes are given in the following table.

	Job 1	Job 2	Job 3	Job 4	Machine Setup
Machine 1	42	70	930	710	10
Machine 2	340	43	120	7	20
Machine 3	560	32	40	9	60
Machine 4	71	760	5	80	85

(a) The company's goal is to minimize the sum of the setup and machine operation times needed to complete all jobs. Formulate this problem as an integer linear program.
Solution: Let $x_{ij} \in \{0,1\}, i, j \in \{1,2,3,4\}$ denote whether job i is assigned to machine j, where 1 means job is assigned. Let $y_j \in \{0,1\},\ j = 1,2,3,4$ indicate whether machine j is used or not (where 1 means it is used) and thus needs setup. For notation simplicity let's denote $p_{ij},\ i, j\ in\{1,2,3,4\}$

processing time of job i on machine j, for example $p_{23} = 32$, and let's denote setup times as $s_1 = 10, s_2 = 20, s_3 = 30$, and $s_4 = 85$.
Now we can model problem as:

$$\begin{aligned}
\min \quad & \sum_{i=1}^{4}\sum_{j=1}^{4} p_{ij}x_{ij} + \sum_{j=1}^{4} s_j y_j \\
\text{s.t.} \quad & \sum_{j=1}^{4} x_{ij} = 1, \ \forall i = 1,2,3,4 \\
& \sum_{i=1}^{4} x_{ij} \leq 4y_j, \ \forall j = 1,2,3,4 \\
& x_{ij} \in 0,1, \ \forall i = 1,2,3,4, \ \forall j 1,2,3,4 \\
& y_j \in \{0,1\}, \ \forall j = 1,2,3,4
\end{aligned}$$

Constraints $\sum_{j=1}^{4} x_{ij} = 1, \ \forall i = 1,2,3,4$ ensure that each job is assigned to only one machine and constraints $\sum_{i=1}^{4} x_{ij} \leq 4y_j, \ \forall j = 1,2,3,4$ ensure that setupis done is machine is used and not done otherwise.

(b) Assume that we want to impose that no more than two jobs can be assigned for any given machine. What constraints should be added to your model to represent this new restriction? (Just write constraint(s) for this part.)

Solution: To ensure that no more than two jobs are assigned to each machine we can add constraint:

$$\sum_{i=1}^{4} x_{ij} \leq 2, \ \forall j = 1,2,3,4$$

(c) Assume that we want to impose the restriction that if machine 1 is used then machine 3 is used. What constraint should be added to your model to represent this new restriction? (Just write constraint(s) for this part.)

Solution: This can be modeled using:

$$y_1 \leq y_3$$

If machine 1 is used then $y_1 = 1$, and above inequality will force $y_3 = 1$. If machine 1 is not used, then $y_1 = 0$ and above inequality will not restrict y_3.

(d) Assume that we want to impose the restriction that either machine 2 or machine 4 is used. What constraint should be added to your model to represent this new restriction? (Just write constraint(s) for this part.)

Solution: This can be modeled as:

$$y_2 + y_4 \geq 1$$

which will ensure that at least one of the machines 2 or 4 is used.

(e) Assume that we want to impose the restriction that: If machine 1 and machine 3 are both set up, then job 1 should not be assigned to machine 3. (Just write constraint(s) for this part.)

Solution: By adding inequalty

$$x_{13} \leq 2 - y_1 - y_3$$

we will ensure that if both machines 1 and 3 are used (i.e. $y_1 = y_3 = 1$) then $x_{13} = 0$ (i.e. job 1 will not be assigned to machine 3). For any other circumatsances above inequality will not restrict variable x_{13}.